

Deliverable 4: Class Diagram & Domain Description

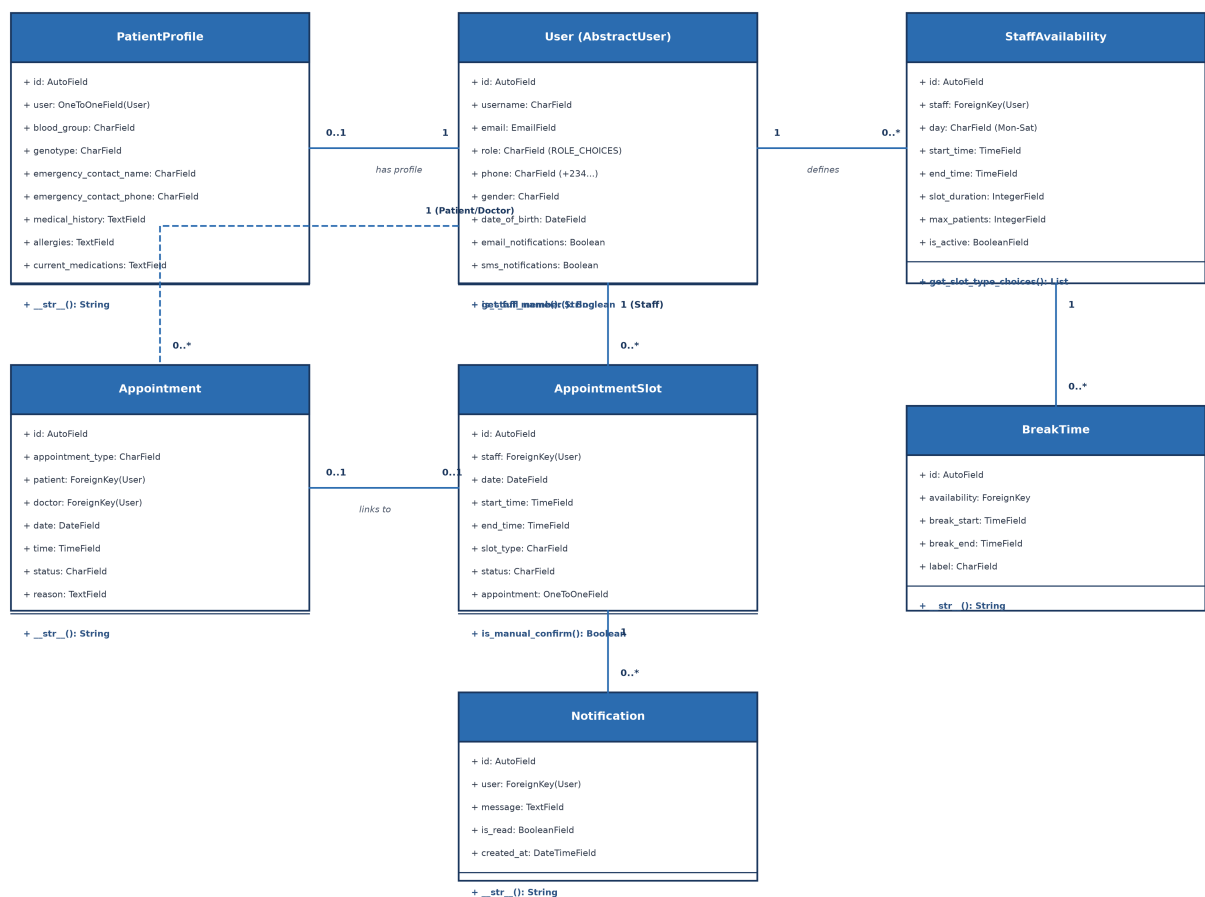
Project: MedSchedule Healthcare Scheduling System | Date: August 2026

1. Object-Oriented Domain Architecture

The MedSchedule Class Diagram illustrates the Django Object-Relational Mapping (ORM) data model. It defines the entity attributes, methods, visibility, cardinalities, and structural relationships (Inheritance, One-to-One, One-to-Many, Composition) governing the scheduling core.

2. System Domain Class Diagram

MedSchedule System - Django Domain Class Diagram



3. Detailed Class Specifications

Class Name	Attributes & Field Types	Key Methods & Relationships
User <i>(AbstractUser)</i>	role: CharField (17 choices) phone: CharField (+234 regex) gender: CharField email_notifications: Bool sms_notifications: Bool	is_staff_member: Property get_full_name(): String Links 1:1 PatientProfile, 1:N Availability
PatientProfile	user: OneToOneField(User) blood_group: CharField genotype: CharField medical_history: TextField allergies: TextField	__str__(): String Stores detailed EHR medical background for patient role.
StaffAvailability	staff: FK(User) day: CharField (Mon-Sat) start_time / end_time: TimeField slot_duration: Integer (default 30)	get_slot_type_choices(): List Composes 1:N BreakTime entities.
BreakTime	availability: FK(StaffAvailability) break_start / break_end: TimeField label: CharField ('Lunch', 'Rest')	Child of StaffAvailability; excludes slot creation during break windows.
AppointmentSlot	staff: FK(User) date: DateField start_time / end_time: TimeField status: CharField (available/booked/etc) appointment: OneToOneField	is_manual_confirm(): Boolean Determines if staff role requires manual review.
Appointment	appointment_type: CharField patient: FK(User) doctor: FK(User) status: CharField (pending/confirmed/etc)	__str__(): String Core booking entity connecting patient and clinical provider.
Notification	user: FK(User) message: TextField is_read: BooleanField created_at: DateTimeField	Stores in-app and log alerts for user dispatches.